BY ORDER OF THE
SECRETARY OF THE AIR FORCE

AIR FORCE INSTRUCTION 11-102

29 MARCH 2002

AIR COMBAT COMMAND
Supplement

7 JUNE 2007

Flying Operations

FLYING HOUR PROGRAM MANAGEMENT

COMPLIANCE WITH THIS PUBLICATION IS MANDATORY

ACCESSIBILITY: Publications and forms are available on the e-Publishing website at **www.e-publishing.af.mil** for downloading or ordering.

RELEASABILITY: There are no releasability restrictions on this publication.

OPR: HQ USAF/XOOT

Certified by: HQ USAF/XOO
(Maj Gen Richard A. Mentemeyer)
Pages: 27

(ACC)

OPR: HQ ACC/A3TB

Certified by: HQ ACC/A3
(Maj Gen (Sel) Douglas L. Raaberg)

Supersedes ACCI 11-103, 16 September 1998

Pages: 9

This instruction implements AFPD 11-1, *Flying Hour Program*, and establishes the Air Force Flying Hour Model. It describes the methodology used to determine the number of MAJCOM flying hours that make up the Air Force Flying Hour Program (FHP). It applies to all USAF flight managers. Send comments and suggested improvements to this instruction on AF Form 847, **Recommendation for Change of Publication,** through channels to HQ USAF/XOOT, 1480 Air Force Pentagon, Washington DC 20330-1480. Major Commands (MAJCOMs) may supplement this instruction. Supplements cannot be less restrictive than the basic publication. MAJCOMs will coordinate supplements to this instruction with HQ USAF/XOOT before publication and will forward one copy to HQ USAF/XOOT after publication. Maintain and dispose of records created as a result of process prescribed in this instruction according to AFMAN 37-139, *Records Disposition Schedule.*

This is a new publication. It must be completely reviewed.

(ACC) AFI 11-102, 5 April 2002, is supplemented as follows. It governs ACC flying hour programs, including development and management; It applies to all Air Combat Command units and subordinate units to which aircraft are assigned. This instruction does not apply to Air National Guard (ANG) or Air Force Reserve Command (AFRC) units or members. Send comments and suggested improvements to this instruction on AF Information Management Tool 847, **Recommendation for Change of Publication**, through channels, to HQ ACC/A3TB, 205 Dodd Blvd, Suite 106, Langley AFB VA 23665-2789, or e-mail to: **acc.dotbb@langley.af.mil**. Maintain official records in accordance with AFMAN 37-123 (to

be AFMAN 33-363), *Management of Records*, and dispose of them in accordance with the AF Records Disposition Schedule found at **https://www.afrims.amc.af.mil**. Contact supporting records managers as required for approval. Reports RCS: A3TB(M)7101, **Aircraft Utilization Report** (AUR B), and AT3B(M)7102, Aircraft Utilization Report (FHOL), are prescribed by this supplement.

SUMMARY OF CHANGES

(ACC) This document is substantially revised and must be completely reviewed. This revision updates guidance and procedures throughout the supplement. It provides field units with detailed standards for planning flying hour programs. Flying hour models can be referenced and downloaded on the HQ ACC/A3TB web page: **wwwmil.acc.af.mil/do/dot/dotb**.

Chapter 1— INTRODUCTION, COMPLIANCE, AND RESPONSIBILITIES 4

1.1. Introduction and Background. 4

1.2. Applicability and Compliance. 4

1.3. Responsibilities 4

1.4. The AFFHM is composed of 5 core components: Force Structure, Aircrew Data, Requirements, Calculation, and Summary (Figure 2.1.). 5

1.5. (Added-ACC) Flying Hour Program (FHP): 5

Chapter 2— THE AIR FORCE SINGLE FLYING HOUR MODEL 7

2.1. Force Structure. 7

Figure 2.1. The Air Force Single Flying Hour Model. 7

2.2. Aircrew Data. 7

2.3. Requirements. 8

2.4. Flying hour computations must include an experiencing (aging) calculation. 10

2.4. (ACC) Program changes and reflows during execution year. 10

2.5. Calculation. 11

2.5. (ACC) ACC/A3TB will determine the number of sorties required 11

2.6. Summary. 12

Attachment 1— GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION 13

Attachment 2— SAMPLE CAF AND MAF FORCE STRUCTURE 16

Attachment 3— SAMPLE AIRCREW DATA 17

Attachment 4— SAMPLE REQUIREMENTS 19

Attachment 5— SAMPLE CALCULATION 21

Attachment 6— SAMPLE SUMMARIES 23

Attachment 7— AIR FORCE SINGLE FLYING HOUR MODEL 25

Attachment 8—(Added-ACC) PROGRAM CHANGE REQUEST: 26

Attachment 9—(Added-ACC) PROGRAM REFLOW OR NEW PROGRAM SUBMISSION 27

Chapter 1

INTRODUCTION, COMPLIANCE, AND RESPONSIBILITIES

1.1. Introduction and Background. The Air Force Flying Hour Program consists of the flying hours necessary to train aircrews to safely operate their aircraft and sustain them in numbers sufficient to execute their core tasked mission. The Air Force Flying Hour Model (AFFHM) provides the methodology and processes that MAJCOMs will use to build their flying hour programs. This model determines the number of flying hours needed to attain and maintain combat readiness for all aircrews, test weapons and tactics, and fulfill collateral requirements. The Joint Mission Essential Task List, the Air Force task lists, and the MDS-specific volumes of the AFI 11-2 series are the foundational requirements that link aircrew training to tasks required to support the warfighting CINCs. The centrality of the flying hour program to the United States Air Force's readiness and combat capability cannot be overemphasized. For this reason it must be defendable and auditable. To that end, it must be standard across the Air Force, be connected to readiness indicators, based on the train-to-task concept, easily understood, and most importantly, be based upon the requirements to train and experience aircrews to perform Air Force missions.

1.1.1. The intent of this instruction is to provide a common methodology and structure for determining flying hour requirements while acknowledging unique MAJCOM requirements. The depiction of the model and the requirements of this instruction are such that they capture the necessary differences in the sortie-based, event-based and throughput-based flying hour programs of the combat, mobility, and formal training forces.

1.2. Applicability and Compliance. Active Duty, Air Force Reserve Command and Air National Guard will use The AFFHM described in this instruction. Flying hour requirements based on operational missions or Research, Development, Test and Evaluation (RDT&E) will comply with the intent of the model.

1.3. Responsibilities . AF/XOOT is the office of primary responsibility for the AFFHM and is the approval authority for suggested changes. MAJCOM and ANG Directors of Operations are responsible for MAJCOM models.

1.3.1. The lead command establishes the training basis for all mission design series (MDS) aircraft in its inventory. User commands must use the same flying hour computations. See AFPD 10-9, *Lead Operating Command Weapon Systems Management*, for further information.

1.3.1. **(ACC)** HQ ACC/A3TB will:

1.3.1.1. **(Added-ACC)** Manage the ACC Flying Hour Program (FHP).

1.3.1.2. **(Added-ACC)** Provide ACC's recommended FHP to HQ USAF in accordance with AFI 11-101.

1.3.1.3. **(Added-ACC)** Coordinate FHPs with other staff agencies for validation.

1.3.1.4. **(Added-ACC)** Validate the Reliability and Maintainability Information System (REMIS) data.

1.3.1.5. **(Added-ACC)** Monitor individual unit flying hour accomplishments on a monthly basis. Manage records created IAW the AF RDS. Contact local records managers as needed.

1.3.1.6. **(Added-ACC)** Coordinate all flying hour program change requests through Air Combat Command Director of Air and Space Operations (ACC/A3) for approval.

1.3.1.7. **(Added-ACC)** Conduct Staff Assistance Visits to each unit as required to be determined by ACC/A3T.

1.3.2. **(Added-ACC)** Flying organizations will:

1.3.2.1. **(Added-ACC)** Operational Support Squadron commander or designee will appoint primary and alternate Flying Hour POCs who will report flying hours to HQ ACC/A3TB. These individuals will be appointed for a minimum of one year and will receive training at the next ACC Flying Hour Symposium.

1.3.2.2. **(Added-ACC)** Ensure flying squadrons plan and execute unit flying hour programs as separate elements of the parent unit's program.

1.3.2.3. **(Added-ACC)** Submit an annual FHP in accordance with paragraph **2.4.1. (Added)**

1.3.2.4. **(Added-ACC)** Submit RCS: A3TB(M) 7101 to HQ ACC/A3TB as early as the 5th but no later than the **10th calendar day of each month via e-mail.**

1.3.2.5. **(Added-ACC)** Submit RCS: A3TB(M) 7102 to HQ ACC/A3TB to report monthly flying hours via Flying Hours on Line (FHOL) **by the 10th calendar day of each month.**

1.3.2.6. **(Added-ACC)** The unit's operations officer will ensure completed AFTO 781's are delivered to the Squadron Aviation Resource Manager (SARM) office not later than the morning after the date of flight. SARM's will validate each day's sorties and hours flown with maintenance debrief personnel daily. SARM personnel will validate daily/weekly AUR's and return correct or adjusted AUR's to maintenance debrief. OSS representatives will validate weekly/monthly totals and ensure FHOL is updated. The purpose of the reviews is to identify and correct errors on post mission paperwork (to include inaccurate time and/or mission symbols) and ensure AFTO Form 781 data input into IMDS, ARMS, and FHOL reflects actual data.

1.3.2.7. **(Added-ACC)** Solicit feedback from HQ ACC/A3TB as required.

1.4. The AFFHM is composed of 5 core components: Force Structure, Aircrew Data, Requirements, Calculation, and Summary (**Figure 2.1.**). For operational flying units the relationship of these components expresses the mathematical description: force structure determines the number of pilots, pilots multiplied by requirements determine the number of flying hours. For formal training units the mathematical description is: average daily student load multiplied by the average number of flying hours per student per day, multiplied by the number of training days determines the number of required student flying hours, which determines force structure. MAJCOMs may add other functions to the model as long as its core structure remains intact.

1.5. (Added-ACC) Flying Hour Program (FHP):

1.5.1. **(Added-ACC)** The ACC FHP represents a considerable fiscal commitment and directly affects the readiness of aircrews in the Combat Air Forces (CAF). Future command flying hour requirements are incorporated into the USAF Program Objective Memorandum (POM) and are included in the USAF Program Document, *Aerospace Vehicles and Flying Hours (PA)*.

1.5.2. **(Added-ACC)** Unit flying hour allocations are derived from a combination of factors to include mission tasking, aircrew training requirements, unit equipment, alert commitment, aircrew ratio, staff requirements, etc. These factors determine a rate of flying necessary to maintain combat capability that is expressed as an aircraft utilization rate (UTE).

1.5.3. **(Added-ACC)** Flying hour goals are cumulative values to be achieved by the end of the fiscal year. The planning and execution of individual monthly programs toward this yearly goal is a unit responsibility. Except for emergencies or Higher Headquarters (HHQ) tasking at year-end (e.g., hurricane evacuations or air sovereignty scrambles), **UNIT FLYING HOUR PROGRAMS WILL NOT BE OVERFLOWN** by more than 20 hours per MDS **WITHOUT PRIOR HHQ APPROVAL**. Unit commanders are not required to "zero out" their annual flying hour program at the end of the fiscal year. The last flying day of the year should be planned and flown as a normal flying day and should not be truncated solely because the annual flying hour contract has been satisfied. Procedures for requesting changes to annual contracts are outlined in paragraph **2.4.2. (Added)** of this supplement.

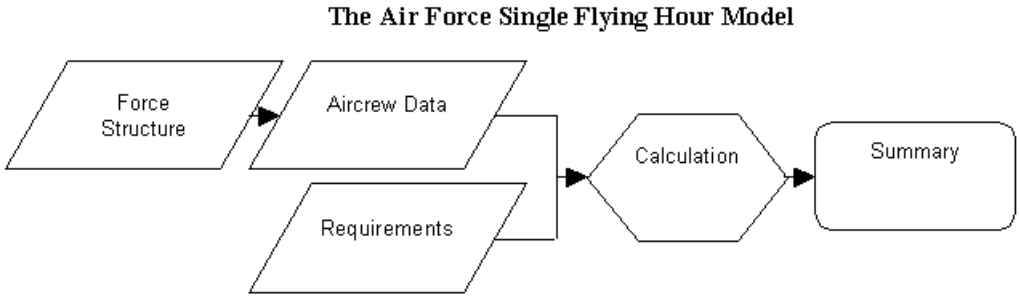
1.5.4. **(Added-ACC)** REMIS is the official USAF flying hour report IAW AFI 21-103, *Equipment Inventory, Status, and Utilization Reporting*; and AFM 66-279V1, *Core Automated Maintenance System (CAMS)*. The importance of accurate and timely flying hour utilization reporting through CAMS and REMIS cannot be overemphasized. Erroneous reporting may result in the loss of required funding, personnel authorizations, and supplies.

Chapter 2

THE AIR FORCE SINGLE FLYING HOUR MODEL

2.1. Force Structure. See **Attachment 2** for examples. This component is the input site for Primary Aircraft Inventory (PAI) and crew ratio, and is used to determine the number of Aircrew Position Indicator (API) 1 pilots. This data may be portrayed by fleet or by unit, at MAJCOM discretion. For pilot production, no input is required because force structure is a function of the student load

Figure 2.1. The Air Force Single Flying Hour Model.



See **Attachment 7** for additional example of Single Flying Hour Model

2.1.2. **(Added-ACC) Aircraft Authorizations:**

2.1.2.1. **(Added-ACC)** HQ USAF specifies the Primary Mission Aircraft Inventory (PMAI) for each ACC unit in the USAF Programs: PA, *Aerospace Vehicles and Flying Hours*; and PD, *Bases, Units, and Priorities*.

2.1.2.2. **(Added-ACC)** Units projected to possess less aircraft than authorized may be assigned a revised PMAI, or Chargeable Aircraft Authorization (CHRG) for Utilization (UTE) and flying hour computations. The CHRG will be displayed for each applicable unit in ACC's allocation messages.

2.2. Aircrew Data. See **Attachment 3** for examples. This component is the input site for the types and number of aircrew members that need to be trained. It includes calculations that result in the number of specific aircrew members (i.e., pilots, navigators, boom operators) that require flying training by position and category. The crew position that drives the greatest number of flying hours is the total requirement. The number of aircrew members is usually derived from crew ratio and PAI in the force structure component. The number of overhead aircrew members is determined by manpower standards. For pilot production, this data is derived from production goals and average daily student load.

2.2.1. Position and category of aircrew members and other pertinent aircrew data include but are not limited to:

2.2.1.1. API 1 Aircraft Commanders, First Pilots and Copilots

2.2.1.2. API 2 Navigators

2.2.1.3. API 6 and 8 staff and supervisory positions at wing level and above

2.2.1.4. Experience mix

2.2.1.5. Instructor Pilots and Supervisors

2.2.1.6. In-unit Requalifications

2.2.1.7. Number of projected upgrades

2.2.1.8. Number of aircrews requiring special qualifications

2.2.1.9. Pilot production goals

2.2.1.10. Average daily student load

2.2.1.11. Number and experience mix of instructor pilots

2.3. Requirements. See **Attachment 4** for examples. This is the input site for the type, number, and/or duration of annual aircrew flying training requirements by aircrew position and category as well as operational mission requirements derived from appropriate tasking documents. Requirements include those events associated with Undergraduate Pilot Training (UPT), initial and mission qualification training, continuation training, upgrade, requalification, and special capability training events/sorties that aircrews must accomplish during the training cycle. Requirements may also include missions performed in support of operational users. Requirement sources include AFI 11-2 MDS and major command Ready Aircrew Program (RAP) messages, and MAJCOM and Numbered Air Force instructions and OPLANs.

2.3.1. Because of mission and training differences, training requirement computations should remain sortie-based for CAF aircraft and event-based for helicopters, airlift and tanker aircraft. In cases where an MDS is operated by different MAJCOMs (C-130Es operated by AMC, PACAF and USAFE), the lead command's methodology takes precedence.

2.3.1. **(ACC)** Flying Hour Program Development and Management.

2.3.1.1. **(Added-ACC)** Budget Activity (BA) Group. Flying hours and funds are assigned to HQ ACC by HQ USAF under two BAs; Operating Forces (BA01), and Administrative and Service-wide Activities (BA04). Within each BA, flying hours are allocated by MDS and Program Element Code (PEC). Each MDS and PEC combination is assigned a separate account managed by ACC and reported to HQ USAF. Hours and funding may not be transferred from one BA to another without HQ USAF approval.

2.3.1.2. **(Added-ACC)** Units with flying hours in more than one PEC will manage each account separately. Only HQ ACC has the authority to transfer flying hours between PECs in the same BA. Units will only use and report flying time under the PECs assigned to them by HQ ACC. Coordination will be established in the above mentioned MOA between the Operations and the CAMS Data Base Managers to ensure the utilization codes are loaded against the correct PEC in the CAMS database.

2.3.1.3. **(Added-ACC)** UTE Requirements. ACC will determine an annual Sortie UTE to each flying hour program. The UTE is expressed in sorties per aircraft per month. The UTE requirement may vary for each unit and MDS depending on each unit's situation.

2.3.1.4. **(Added-ACC)** Average Sortie Duration (ASD). ASD is used to convert sorties to flying hours and vice versa. HQ ACC/A3TB uses the unit's last programmed ASD when initially determining flying hour programs for the current and future years. Units will update their ASDs annually to reflect the unit's best estimate of the optimum sortie duration after considering historical experience, changes in missions or mix of missions, distance to ranges, deployments, etc.

2.3.1.5. **(Added-ACC)** Initial Program Submission. By the end of May HQ ACC/A3 will provide the units a projection for the upcoming fiscal year inputs through a "Proposed FHP" message.. HQ ACC/A3TB will formulate the next year's projected allocation for each unit and MDS using information from HQ ACC staff agencies, historical unit flying data, the ACC Flying Hour Program Model and unit-specific inputs. Additionally, information such as Programmed Flight Training (PFT), staff training, and test requirements, will be identified and included in the initial program build. Formal Training Units (FTUs), test units, and other ACC aircraft not under the Ready Aircrew Program (RAP) will project their flying hour programs based on historical data, PFT, test schedules, etc.

2.3.1.5.1. **(Added-ACC)** Upon receipt of the Proposed FHP, units will review and provide HQ ACC/A3TB with any projected changes and rationale for the change. Units will respond by submitting total requested sorties/hours for their annual flying hour programs to ACC/A3TB. Units submitting projected sorties/hours more than one percent in variance from the ACC/A3 "Proposed FHP" message will provide documentation staffed through Wing OG. Units unable to meet Proposed FHP flying hour projections due to maintenance concerns or limitations will submit documentation coordinated through ACC/A4.

2.3.1.5.2. **(Added-ACC)** Upon completion and review of all unit responses to the "Proposed FHP" message, HQ ACC/A3TB will submit a proposed flying hour program for COMACC approval. HQ ACC/A3 will provide each unit with its final tasking for the next fiscal year in the "Baseline Allocation" message. Units receiving baseline allocations different from their flying hour proposal should submit a flow plan based on the new program to HQ ACC/A3TB prior to the beginning of the fiscal year.

2.3.1.6. **(Added-ACC)** Contingency Planning. Since the flying hours and associated funds provided to ACC by Congress through HQ USAF are allocated for aircrew training, maintenance of combat capability, test functions, and higher headquarters tasking; contingency operations may require additional funding by national command authorities. Because there is some inherent training that occurs while participating in contingency operations, the actual funding and programming for contingency operations requires resources allocated to support the unit's home station flying hour program along with additional flying hours/funds supplemented by the HQ USAF.

2.3.1.6.1. **(Added-ACC)** The unit's contribution to the contingency operation will be those hours the deployed aircraft/crew ratio would have flown at home station in their normal training profiles. HQ ACC/A3TB will reprogram the hours from the home station training account and place them into a specific contingency account for the unit to fly. Due to higher ASDs in theater, this account will normally have fewer hours than the total hours required, necessitating contingency supplemental hours from HQ USAF.

2.3.1.6.2. **(Added-ACC)** Deployed units will fly all tasking assigned by the theater commander. HQ ACC/A3TB is responsible for ensuring the unit obtains the flying hours required to fulfill theater requirements and will supplement the account with additional flying hours, by authority received from HQ USAF. However, to ensure the optimum use of contingency supplemental flying hours and dollars, the unit must provide accurate and timely execution data from their deployed forces. Units execute the hours in their contingency operation account beginning when their aircraft takeoff from home station deploying to the theater and continue to use them until their aircraft land back at home station on redeployment. Units are not accountable for all the hours that are in their contingency accounts.

2.3.2. Examples include:

2.3.2.1. Ready Aircrew Program (RAP) Sorties (those sorties that lead to basic and combat mission skills) and non-RAP sorties (those sorties that build basic pilot skills such as instrument, advanced handling, navigation, etc.)

2.3.2.2. Mission Qualification Training

2.3.2.3. Special Capability Requirements

2.3.2.4. Operational Missions

2.3.2.5. Collateral and other force support sorties/hours (ferry, functional check flight, weather ship, control ship, etc.)

2.3.2.6. Aging rate required for aircrews to achieve required crew qualifications

2.3.2.7. Number of training events

2.3.2.8. In-unit requalification training

2.3.2.9. Syllabus hours associated with undergraduate and graduate flying training

2.3.2.10. Refly rate, scheduling effectiveness/efficiency

2.3.2.11. Number training days

2.3.2.12. Flying Hour Factor (FHF, the average number of flying hours per student per day) x refly rate

2.4. Flying hour computations must include an experiencing (aging) calculation. Although the terms are different for fighter versus multi-place, crewed aircraft, copilots and wingmen must accumulate hours permitting them to upgrade at a minimal rate to support planned absorption and crew qualification requirements to maintain a unit's ability to fulfill its assigned missions. This calculation will ensure that flying hour programs identify and provide the required hours to upgrade at a prescribed rate and ensure a standardized requirements computation for all aircraft.

2.4. (ACC) Program changes and reflows during execution year. Program Change requests to unit flying hour programs will only be submitted when the unit's total flying hour requirement changes for the year. Program changes should be made from the unit Operations Group Commander or higher and should be addressed to HQ ACC/A3T. HQ ACC/A3TB will staff the request for approval. Requested changes must be fully justified. Units are required to submit a memo for flying hour change approval (**Attachment 8 (Added)**) and a revised program reflow (**Attachment 9 (Added)**).

NOTE: Changes to a unit's program constitute a new "contract" with the unit. Units requesting a change to their annual flying hour program will receive a confirmation response once the proposed change is approved.

2.4.1. **(Added-ACC)** Adjustments to flying hour programs affect Operations and Maintenance (O & M) funding at the unit. Because unit commanders have the fiduciary responsibility for all activities at their unit, all flying hour related O&M dollars are allocated to the unit level. Therefore, a unit requiring additional flying hours for additional training requirements will require the O&M dollars to support the hours. Conversely, if a unit does not intend to fly all their flying hours, the funds to support them are not required at that unit. ACC policy is O&M dollars follow hours, and all changes to a unit's annual flying hour program will be followed by a corresponding change to the unit's O&M funding. In

cases where actual costs per flying hour exceed funded levels (due to increased spare parts usage, etc.), only the difference between the allocation funding and the amount absorbed by the unit for the increased costs, if any, will be withdrawn. Under executed hours or funding will not be withdrawn until the middle of the fourth quarter or later in order to give units every opportunity to fly the program.

2.4.2. **(Added-ACC)** A program reflow of a unit's flying hour program is basically the same as developing the yearly program. Additionally, CHRG aircraft should remain constant for the months already flown. The remaining hours and sorties are allocated month by month for the remainder of the year. Computations are based on unit needs; adjusting sorties, flying hours, ASD, and UTE as required ensuring the net total hours already flown plus the additional requirement equals the unit's total flying hour "contract". For example, a revision requested in May must show the actual data flown from Oct - Apr and re-computed data from May - Sep. This request must be formatted as shown in **Attachment 9 (Added)** and received NLT the 10th calendar day of the month to be applicable for that month. The main reason for a re-flow will be due to a change in contingency tasking.

2.5. Calculation. See **Attachment 5** for examples. Flying hour requirements are based on the number of aircrew members that need to be trained and their annual flying training requirements. The following basic formula applies:

2.5. (ACC) ACC/A3TB will determine the number of sorties required to train aircrews to COMACC-directed readiness levels and maintain aircrew manning and experience levels (AFI 11-2MDS volume sets and RAP tasking). Total sorties / 12 months / Primary Aircraft Authorized (PAA) = Sortie UTE Rate (SUTE). Sorties X ASD = Flying Hours.

2.5.1. For operational flying units: Hours = number aircrews by category x requirements x duration x refly

2.5.2. For pilot production units: Hours = FHF x class load x number of training days. Instructor pilot continuation training requirements are determined in the same manner as operational pilots.

2.5.3. Within this area the individual formulas are listed that calculate the hours necessary to meet each individual training requirement. In general, each requirement will be represented by its own formula that yields flying hours specific to that requirement. Examples of operational training formulas include:

2.5.3.1. Combat Mission Ready hours (separated for experienced and inexperienced pilots)

2.5.3.2. Basic Mission Qualified hours (separated for experienced and inexperienced pilots)

2.5.3.3. Navigation Training/Instrument/Advanced Handling Characteristics hours

2.5.3.4. Hours required to meet additional special missions/capabilities specifically tasked to the unit, such as Combat Search and Rescue, AGM-88, AGM-130, etc.

2.5.3.5. Hours necessary to maintain Instructor Pilot/ Supervisor qualification

2.5.3.6. Hours necessary to conduct Mission Qualification Training

2.5.3.7. Hours required to account for scheduling efficiency and refly requirements (these are historically based)

2.5.3.8. Hours associated with take-off (initial and non-initial), landings, cell formations, air refueling (receiver and tanker)

2.5.3.9. Overseas requirements

2.5.3.10. Aging or upgrade requirements

2.5.4. Examples for Formal Training formulas include:

2.5.4.1. For each MDS, calculations producing the number of training days by month and class, number of students by month and class,

2.5.4.2. Lastly, a position summarizing by MDS indicating total student, instructor pilot continuation training, and collateral flying hours

2.6. Summary. See **Attachment 6** for examples. This is a display of the resulting flying hours. MAJCOMs have the latitude to format this display to meet their needs but as a minimum it will show flying hours by MDS.

CHARLES F. WALD, Lt General, USAF
DCS/Air & Space Operations

(ACC)

RONALD E. KEYS, General, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFPD 10-9, *Lead Operating Command Weapon Systems Management*

AFPD 11-1, *Flying Hour Program*

AFMAN 37-139, *Records Disposition Schedule.*

Abbreviations and Acronyms

AFFHM—Air Force Flying Hour Model

AMC—Air Mobility Command

ANG—Air National Guard

API—Aircrew Position Indicator

BMC—Basic Mission Capable

CAF—Combat Air Forces

CMR—Combat Mission ready

FHF—Flying Hour Factor

FHP—Flying Hour Program

MAF—Mobility Air Forces

MAJCOM—Major Command

MDS—Mission Design Series

MQT—Mission Qualification Training

PACAF—Pacific Air Forces

PAI—Primary Aircraft Inventory

RAP—Ready Aircrew Program

RDT&E—Research Design Test and Evaluation

USAFE—US Air Forces in Europe

(ACC) Abbreviations and Acronyms

ACC—Air Combat Command

AFRC—Air Force Reserve Command

API—Aircrew Position Indicator

ASD—Average Sortie Duration

AUR—Aircraft Utilization Report

BA—Budget Activity

CAF—Combat Air Forces

CAMS—Core Automated Maintenance System

FHOL—Flying Hours On Line

FHP—Flying Hour Program

FTU—Formal Training Unit

GCC—Graduated Combat Capability

O&M—Operations and Maintenance

OSS—Operations Support Squadron

PEC—Program Element Code

PMAI—Primary Mission Aircraft Inventory

POM—Program Objective Memorandum

REMIS—Reliability and Maintainability Information System

SCM—Sorties per Crew per Month

USAF—United States Air Force

UTE—Utilization

(ACC) Terms

Backup Aircraft Inventory (BAI)—Aircraft above the PMAI to permit scheduled and unscheduled maintenance, modifications, inspections and repair without reduction of aircraft available for operational missions.

Change—A recompilation of a unit's month-by-month flying hour plan, this is required when the unit's flying hour allocation changes.

Chargeable Aircraft (CHRG ACFT)—The number of aircraft against which units should build their programs. Except in cases where possessed aircraft is forecast to be significantly different from the PMAI such as in building or down sizing units, chargeable aircraft will normally equal the unit's PMAI, PTAI, or PDAI, as applicable. In these cases, HQ ACC/DO/LG will assign a chargeable aircraft accountability for the unit in the ACC Flying Hour First Look and Baseline Messages.

Mission, Design, and Series (MDS)—An acronym for aircraft mission, design, and series. For example: B052H, F015C, etc.

Possessed Aircraft—Aircraft under a wing commander's operational control and responsibility IAW AFI 21-103.

Primary Aircraft Inventory (PAI)—Aircraft assigned to meet the primary aircraft authorization (includes PDAI, PMAI, POAI, PTAI). PMAI will not change except when approved by HQ USAF.

Primary Development/Test Aircraft Inventory (PDAI)—*Formerly CB or Test.* Aircraft assigned primarily for the test of the aircraft or its components for purposes of research, development, test and evaluation, operational test and evaluation, or support for testing programs.

Primary Mission Aircraft Inventory (PMAI)—*Formerly CC/CA PAA Coded Aircraft*. Aircraft assigned to a unit for performance of its wartime mission.

Primary Other Aircraft Inventory (POAI)—*Formerly ZA, ZB*. Aircraft required for special missions not elsewhere classified.

Primary Training Aircraft Inventory (PTAI)—*Formerly TF*. Aircraft required primarily for technical and specialized training for crew personnel or leading to aircrew qualification.

Program Element (PE)—The PE is the smallest unit of military output controlled at the DOD level. It is identified by a six-digit alphanumeric program element code (PEC). The sixth character, “F", identifies the PE with the Air Force.

Program Element Code (PEC)—The six digit alphanumeric code used to identify the Program Element (see definition above).

Ready Aircrew Program (RAP)—Continuation Training regulated under the AFI 11-2MDS volume sets for aircrews assigned to units primarily flying fighter, bomber, and LDHD PMAI. The ACC flying hour program centers around unit RAP taskings and the associated flying hours derived using the flying hour program models.

Reflow—An update of a unit's month-by-month flying hour plan, this is required when the unit changes its monthly flow plan without changing allocated hours.

Total Active Inventory (TAI)—Aircraft assigned to operating forces for mission, training, test, or maintenance functions (includes primary aircraft inventory, backup aircraft inventory, attrition, and reconstitution reserve).

UTE Remaining—A measurement of the UTE required to accomplish a unit's remaining flying hours with assigned aircraft over the remaining months of the fiscal year.

Utilization Rate (UTE)—For ACC aircraft, the UTE is expressed in the number of Sorties flown per Aircraft per Month.

Attachment 2

SAMPLE CAF AND MAF FORCE STRUCTURE

Figure A2.1. Sample CAF Force Structure.

Force Structure	FY 02	FY 03	FY 04	FY 05	FY 06
PAA	24	24	24	24	24
Crew Ratio	1.25	1.25	1.25	1.25	1.25

Figure A2.2. Sample MAF Force Structure.

MDS: KC-135 Crew Ratio: 1.36

Squadron PAA	FY 02	FY 03	FY 04	FY 05	FY 06
XX ARS Qtr 1	12	12	12	12	12
XX ARS Qtr 2	12	12	12	12	12
XX ARS Qtr 3	12	12	12	12	12
XX ARS Qtr 4	12	12	12	12	12
XX ARS FY Avg	12	12	12	12	12
API-1 Crews	FY 02	FY 03	FY 04	FY 05	FY 06
XX ARS Qtr 1	17	17	17	17	17
XX ARS Qtr 2	17	17	17	17	17
XX ARS Qtr 3	17	17	17	17	17
XX ARS Qtr 4	17	17	17	17	17
AA ARS FY Avg	17	17	17	17	17

Attachment 3

SAMPLE AIRCREW DATA

Figure A3.1. Sample Aircrew Data (Sortie-Based MDS).

Aircrew Data	FY 02	FY 03	FY 04	FY 05	FY 06
Aircrews					
API 1 Experienced (50%)	12	12	12	12	12
API 1 Inexperienced (50%)	12	12	12	12	12
API 6 CMR Experienced	4	4	4	4	4
API 6 CMR Inexperienced	0	0	0	0	0
API 6 BMC Experienced	4	4	4	4	4
API 6 BMC Inexperienced	0	0	0	0	0
API 8 Experienced	2	2	2	2	2
API 8 Inexperienced	0	0	0	0	0
Total Crews	34	34	34	34	34
Additional Requirements					
IP/Supervisors	8	8	8	8	8
CSAR Experienced	4	4	4	4	4
CSAR Inexperienced	4	4	4	4	4
AGM-88	12	12	12	12	12
MQT Students	21	21	21	21	21
Advisors (AFRC)					

Figure A3.2. Sample Aircrew Data (Events Based).

Aircrew Factors	FY 02	FY 03	FY 04	FY 05	FY 06
CP per AC per crew	1	1	1	1	1
No. of Aircraft Commanders	238	238	238	238	238
No. of CP/FP	238	238	238	238	238
API 1 Receiver Aircrews	16	16	16	16	16
API 6 Staff Positions	85	85	85	85	85
API 6 Staff Receiver Pilots	6	6	6	6	6
API 8 Staff Positions	15	15	15	15	15
Instructor Pilot Upgrades %	33.3	33.3	33.3	33.3	33.3
Instructor Pilot Upgrades	80	80	80	80	80
Aircraft Commander Local Upgrades %	20	20	20	20	20
Aircraft Commander Local Upgrades	24	24	24	24	24

Attachment 4

SAMPLE REQUIREMENTS

Figure A4.1. Sample Requirements (Sortie-Based).

	FY 02	FY 03	FY 04	FY 05	FY 06
Force Sustainment Sorties	4428	4428	4428	4428	4428
CMR Experienced Sorties	1440	1440	1440	1440	1440
CMR Inexperienced Sorties	1392	1392	1392	1392	1392
BMR Experienced Sorties	360	360	360	360	360
BMC Inexperienced Sorties	0	0	0	0	0
Nav Tng Sorties	162	162	162	162	162
INST/AHC Sorties	198	198	198	198	198
MQT Sorties	189	189	189	189	189
CSAR Exp Sorties	32	32	32	32	32
CSAR Inexp Sorties	48	48	48	48	48
AGM-88 Sorties	48	48	48	48	48
IP/Supervisor Sorties	96	96	96	96	96
Scheduling Efficiency Sorties	417	417	417	417	417
Attrition Sorties	46	46	46	46	46

Special Cap. Sorties	FY 02	FY 03	FY 04	FY 05	FY 06
Nav Tng Sorties	6	6	6	6	6
Inst/AHC Sorties	6	6	6	6	6
MQT Sorties	9	9	9	9	9
CSAR Exp	8	8	8	8	8
CSAR Inexp	12	12	12	12	12
AGM-88	4	4	4	4	4
IP/Supervisor Sorties	12	12	12	12	12

Figure A4.2. Sample Requirements (Events-Based).

	FY 02	FY 03	FY 04	FY 05	FY 06
Initial Takeoff:					
Events Required for API 1 ACs	**10**	**10**	**10**	**10**	**10**
Events Required for API 1 CPs	**10**	**10**	**10**	**10**	**10**
Events Required for API 6 ACs	**4**	**4**	**4**	**4**	**4**
Events Required for API 8 ACs	**4**	**4**	**4**	**4**	**4**
Event Duration (hours)	**0.5**	**0.5**	**0.5**	**0.5**	**0.5**
Instrument Approach:					
Events Required for API 1 ACs	**46**	**46**	**46**	**46**	**46**
Events Required for API 1 CPs	**46**	**46**	**46**	**46**	**46**
Events Required for API 6 ACs	**30**	**30**	**30**	**30**	**30**
Events Required for API 8 ACs	**12**	**12**	**12**	**12**	**12**
Event Duration (hours)	**0.3**	**0.3**	**0.3**	**0.3**	**0.3**

Upgrade Requirements	**FY 02**	**FY 03**	**FY 04**	**FY 05**	**FY 06**
Instructor pilot upgrade requirements as percentage of API 1 ACs	**33.3**	**33.3**	**33.3**	**33.3**	**33.3**
Pre-CIFC Requirements	**10**	**10**	**10**	**10**	**10**
Average pre-CFIC sortie duration (hrs)	**4.0**	**4.0**	**4.0**	**4.0**	**4.0**
Pre-CFIC sortie requirements	**3**	**3**	**3**	**3**	**3**
In-Unit AC Upgrade Requirements					
Percentage of local upgrades	**20**	**20**	**20**	**20**	**20**
Percentage of API-1 CPs upgrading	**50**	**50**	**50**	**50**	**50**
Average upgrade sortie duration (hrs)	**4.0**	**4.0**	**4.0**	**4.0**	**4.0**
In-unit AC upgrade sortie requirements	**7**	**7**	**7**	**7**	**7**
Average AC upgrade evaluation sortie duration	**4.0**	**4.0**	**4.0**	**4.0**	**4.0**
In-unit AC upgrade evaluation sortie requirements	**1**	**1**	**1**	**1**	**1**

Attachment 5

SAMPLE CALCULATION

Figure A5.1. Sample Calculation.

Calculation	FY 02
Total Hours	6294
Force Sustainment Hours	6022
CMR Exp Hours	1958.4
CMR Inexp Hours	1893.1
BMC Exp Hours	489.6
BMC Inexp Hours	0
Nav Tng Hours	220.3
INST/AHC Hours	269.3
MQT Hours	257.0
CSAR Exp Hours	43.5
CSAR Inexp Hours	65.3
AGM-88 Hours	65.3
IP/Supervisor Hours	130.6
Scheduling Efficiency Sorties	567.1
Refly Sorties	62.6
Force Support Hours	272.0
Aircraft ASD	1.36

Event Driven Training	FY 02
Line Crews (API-1)	
Initial Takeoff (P010)	2,380
Non-initial Takeoff (P020)	1,523
Instrument Approach (P070)	6,569

Landing (P190)	**1,999**
Cell Formation (F020)	**595**
Receiver Air Refueling (R010)	**176**
Receiver Transition	**211**
Tanker Air Refueling (R060)	**7,497**
Air Refueling Transition	**5,988**
Overseas (M030)	**7,140**
Overseas Credit	**- 714**
Unit-Specific Training Sortie (M020)	**11,424**
SUBTOTAL	**44,798**

Aging Requirement	**FY 02**
Inexperienced Pilots	**205**
Experienced Pilots	**61,500**
Non-experienced Hours	**1,842**
PROGRAM TOTAL	**63,342**

Attachment 6

SAMPLE SUMMARIES

Figure A6.1. F-16 Sample Summary.

F-16 Summary	FY 02	FY 03	FY 04	FY 05	FY 06
Total Hours	12588	12588	12588	12588	12588
Force Sustainment	12044	12044	12044	12044	12044
Force Support	544	544	544	544	544
Unit Hours					
Unit A					
Force Sustainment	6022	6022	6022	6022	6022
Force Support	272	272	272	272	272
Unit B					
Force Sustainment	6022	6022	6022	6022	6022
Force Support	272	272	272	272	272

Figure A6.2. C-141 Sample Summary.

C-141	FY 02	FY 03	FY 04	FY 05	FY 06
Force Production	1,728	1,728	1,728	1,728	1,728
Force Sustainment	61,342	61,342	61,342	61,342	61,342
Force Support	272	272	272	272	272
User Support	0	0	0	0	0
TOTAL	63,342	63,342	63,342	63,342	63,342

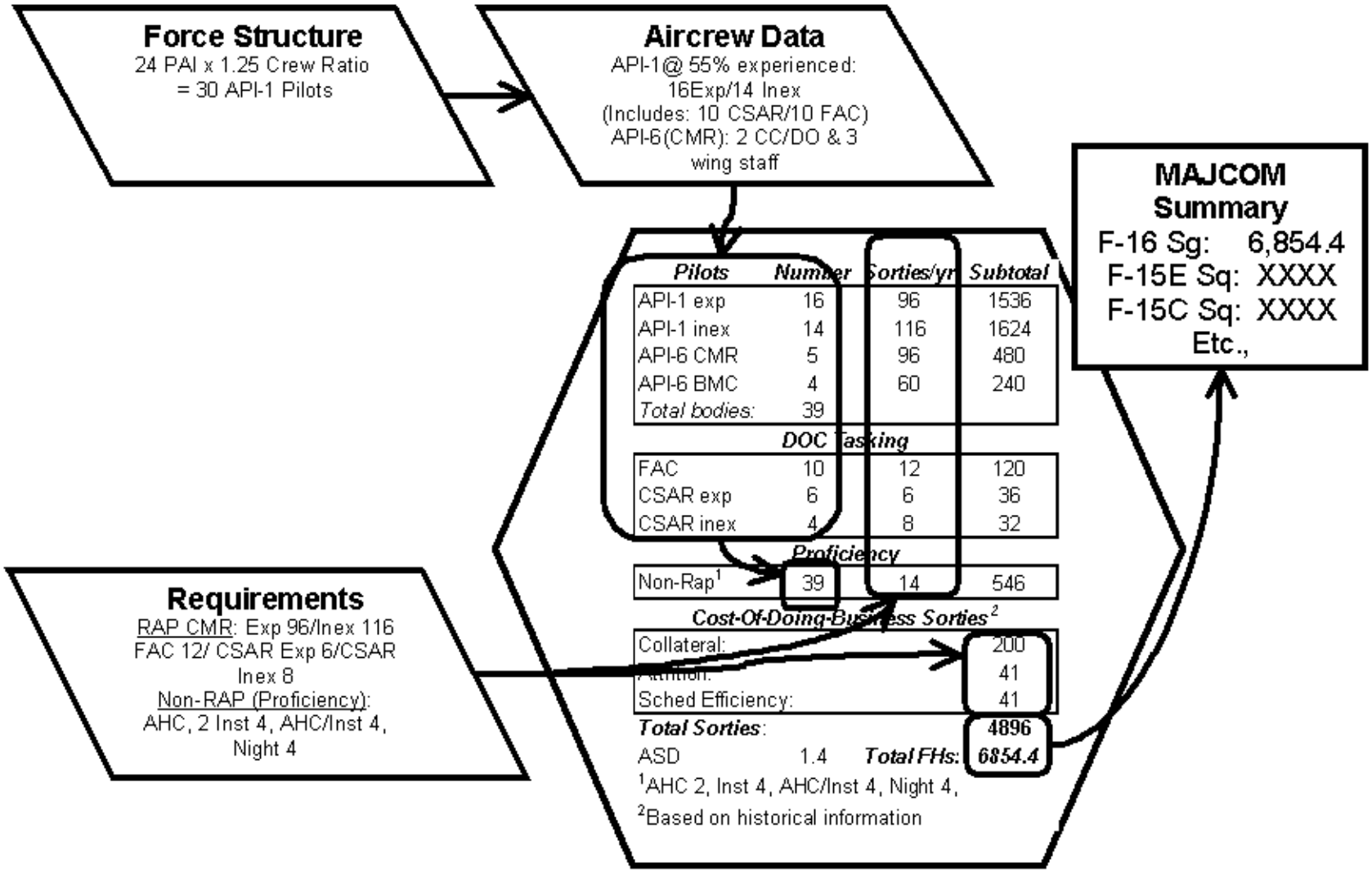
Figure A6.3. T-37 Flying Hour Summary.

Force Production Summary	FY 01	FY 02	FY 03	FY 04	FY 05
Student Hours (SUPT)	40983	34999	24142	11285	0
Student Hours (ALP)	1600	4250	1770	0	0
Student Hours (FWQ)	0	0	0	0	0
Student Hours (Other)	0	0	0	0	0
TOTAL Student Hours	42583	39249	25912	11285	0
Force Sustainment					
Experienced	3541	3145	2120	722	0
Inexperienced	1926	1711	1153	393	0
AFRC CT Additive	1014	820	563	192	0
TOTAL	6481	5675	3836	1307	0
Force Support					
Mission Support	793	793	793	793	0
PAA Rounding	507	267	115	301	0
MX Support	509	464	310	138	0
Support Other					
TOTAL	1808	1524	1218	1232	0
PROGRAM TOTAL	50872	46449	30966	13824	0
PAI	92	84	56	25	0

Attachment 7

AIR FORCE SINGLE FLYING HOUR MODEL

Figure A7.1. Air Force Single Flying Hour Model (Example).



Attachment 8 (Added-ACC)

PROGRAM CHANGE REQUEST:

MEMORANDUM FOR ACC/A3T

FROM: Unit OG/CC

SUBJECT: XX Wing Flying Hour Program Change # XX

1. XX Wing requests a (decrease/increase of XXXX sorties and XXXX hours for the FY XX flying hour program.

2. Changes are as follows (ensure each MDS and mission are listed separately):

MDS	Mission	PEC	Old Program	New Program	Delta
X-XXX	Ops	XX	Srts/Hrs	Srts/Hrs	Srts/Hrs

3. Monthly reflow starting from the beginning of the year (**Attachment 9 (Added)**)

4. "Justification for the Change"

5. This is an Operations/Logistics coordinated memo.

6. POC is XXXXX, DSN XXX-XXXX, E-Mail xxxxxxxxxxx.

Attachment 9 (Added-ACC)

PROGRAM REFLOW OR NEW PROGRAM SUBMISSION

MEMORANDUM FOR ACC/A3T

FROM: Unit OG/CC

SUBJECT: XX Wing Flying Hour Reflow # XX of Program Change Request # XX

1. Reflows are as follows (Ensure each MDS and mission are listed separately):

2.

MDS	PEC	MISSION
X-XXX	XXXXX	Ops/Tng/Test

MONTH	ACFT	SORTIES	ASD	UTE HOURS

(List months Oct - Sep with appropriate data and include total row at the bottom)

3. This is an Operations/Logistics coordinated memo.

4. POC is XXXXX, DSN XXX-XXXX, E-Mail xxxxxxxxxxx.